

B.Tech II Year II Semester (R20) Supplementary Examinations November 2024

LINEAR AND DIGITAL IC APPLICATIONS

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Mention the advantages of integrated circuits. | 2M |
| (b) What is slew rate? Define thermal drift? | 2M |
| (c) Draw the op amp adder circuit. | 2M |
| (d) Draw the circuit of op amp Square waveform generator. | 2M |
| (e) Why active filters are preferred? | 2M |
| (f) List the applications of 565 PLL. | 2M |
| (g) Give the drawbacks of weighted resistor type DAC. | 2M |
| (h) What is the significance of Switching Regulator? | 2M |
| (i) Which of the parameters decide the fan-out and how? | 2M |
| (j) State the effect of loading CMOS output. | 2M |

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Discuss the classification of IC's. 5M
- (b) Explain Dominant pole frequency compensation in op-amp. 5M

OR

- 3 (a) Explain the following terms in an OP-AMP. 5M
- (i) Input Bias current, (ii) Input offset, (iii) Input offset current.
- (b) Write a technical note on frequency response characteristics of differential amplifier. State the importance of frequency compensation. 5M
- 4 (a) Explain practical integrator circuit using IC 741. 5M
- (b) Explain Log and Antilog amplifier with neat sketch. 5M

OR

- 5 What is instrumentation amplifier? What are the required parameters of an instrumentation amplifier? Explain the working of instrumentation amplifier with neat circuit diagram. 10M
- 6 (a) Draw the circuit of a 1st order band pass filter and derive its transfer function. 5M
- (b) Derive the voltage to frequency converter factor for VCO. 5M

OR

- 7 (a) Draw the circuit of all pass filter and derive its transfer function. 5M
- (b) Draw the block diagram of monostable multivibrator using 555 timer and derive an expression for its frequency of oscillation. 5M
- 8 (a) Explain the internal structure of voltage regulator IC 723. 5M
- (b) Write the specifications of ADC. 5M

OR**Contd. in Page 2**

- 9 (a) With neat diagram, explain the working principle of R-2R ladder type DAC. 5M
(b) Write about IC voltage Regulators. 5M
- 10 (a) Draw the resistive model of a CMOS inverter circuit and explain its behavior for LOW and HIGH outputs. 5M
(b) Explain the operation of encoders. 5M
- OR**
- 11 (a) Design 16 bit adder using two 7483 ICs. 5M
(b) Explain magnitude comparators. 5M

B.Tech II Year II Semester (R20) Regular & Supplementary Examinations April/May 2024

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(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|--|----|
| (a) Give the Classification of IC's. | 2M |
| (b) Compare and contrast an ideal op-amp and practical op-amp. | 2M |
| (c) Derive the expression for voltage gain of non-inverting amplifier. | 2M |
| (d) Write the applications of V-I Converter. | 2M |
| (e) What is the need of higher order filters? Give the relationship between order of a filter and roll off rate. | 2M |
| (f) Define capture range and lock in range of a PLL. | 2M |
| (g) Define the terms related to ADC (i) conversion time, (ii) Percentage Resolution. | 2M |
| (h) What are the different features of IC 723? | 2M |
| (i) Write about basic CMOS Inverter. | 2M |
| (j) Write about Universal Shift Register. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | |
|--|----|
| 2 (a) Explain internal circuit of operational amplifier with different stages. | 5M |
| (b) Discuss the AC characteristics of an op-amp in detail. | 5M |

OR

- | | |
|---|----|
| 3 (a) Draw and explain the equivalent circuit of an operational amplifier. Give its features. | 5M |
| (b) Explain about DC Characteristics of an Op-Amp. | 5M |

- | | |
|--|----|
| 4 (a) Derive the output voltage V_O of practical integrator circuit. | 5M |
| (b) Explain the square wave generator using an op-amp with a neat circuit diagram. | 5M |

OR

- | | |
|---|----|
| 5 (a) Explain working of an instrumentation amplifier with a neat diagram. How is it different from a single op-amp differential amplifier? | 5M |
| (b) Explain the operation of Op-Amp based comparator with a neat diagram. | 5M |

- | | |
|---|----|
| 6 (a) Explain about band reject and all Pass filters. | 5M |
| (b) Explain about voltage controlled Oscillator. | 5M |

OR

- | | |
|---|----|
| 7 (a) Compute the free running frequency f_o , lock in range and capture range of PLL565. Assume $R_T = 20\text{ K}\Omega$, $C_T = 0.01\mu\text{F}$, $C = 1\mu\text{F}$ and supply voltage is $\pm 6\text{V}$. | 5M |
| (b) Draw the circuit of Schmitt trigger using IC555 timer and explain its operation. | 5M |

- | | |
|---|----|
| 8 (a) Discuss about the three terminal adjustable voltage regulator and write its applications. | 5M |
| (b) Explain the working of R-2R ladder DAC with a circuit diagram and write its limitations. | 5M |

OR

Contd. in Page 2

- 9 (a) Discuss about Op-Amp based series voltage regulator and write its purpose. 5M
(b) Explain the working of dual slope ADC with a circuit diagram and write its applications. 5M
- 10 (a) Design and explain the function of a CMOS transistor circuit for 3-input AND gate. 5M
(b) What is multiplexer? Draw the logic diagram of 8 to 1 line multiplexer. 5M
- OR**
- 11 (a) Draw the circuit of MOD 16 Down ripple counter with D-flip-flops & explain its function. 5M
(b) With a neat sketch explain the operation of four-bit parallel adder. 5M

B.Tech II Year II Semester (R20) Regular & Supplementary Examinations August/September 2023

LINEAR AND DIGITAL IC APPLICATIONS

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|--|----|
| (a) State the characteristics of ideal op-amp. | 2M |
| (b) Draw the mono stable multivibrator using IC 555. | 2M |
| (c) Give the conversion time for successive approximation ADC. | 2M |
| (d) Give the important features of an instrumentation amplifier. | 2M |
| (e) Define unity gain band width of an op-amp. | 2M |
| (f) Which type of ADC is the fastest? Why? | 2M |
| (g) Realize EX-OR gate with CMOS circuit. | 2M |
| (h) Define slew rate. What causes it? | 2M |
| (i) Mention any two applications of multiplier IC. | 2M |
| (j) Compare R-2R and weighted resistor types of DACs. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Draw Block diagram of Typical Op–Amp With Various Stages and explain in detail. 10M
- OR**
- 3 With the circuit diagram explain the working of Instrumentation Amplifier. 10M
- 4 Derive expression for the voltage to frequency conversion factor of a VCO. 10M
- OR**
- 5 Design an Astable Multivibrator having an output frequency 15 KHz with duty cycle of 40%. 10M
- 6 Design and explain the operation of All Pass Filter with its characteristics? 10M
- OR**
- 7 Derive expression for the duty cycle of Astable Multivibrator using IC 555 with a neat circuit and waveforms. 10M
- 8 Explain the characteristics of three terminal IC Voltage regulator IC273. Design an Voltage regulator using the above IC. 10M
- OR**
- 9 Explain the operation of parallel comparator type ADC. 10M
- 10 Design and implement the function $Z = A(B+C)$ in CMOS logic. 10M
- OR**
- 11 Draw the logic diagram equivalent to the internal structure of an 8-input CMOS NAND gate. Show the transistor circuit for this gate and explain the operation with the help of function table. 10M
